

Appendix C: Observational Anisotropy Maps and Lensing Signatures

C.1 Theoretical Motivation: The BMI Interpretation of Cosmic Strings

While standard cosmological models treat cosmic strings as arbitrary topological defects—often attributed to hypothetical phase transitions in the early universe—Bimodal Manifold Interaction (BMI) Theory derives their existence as a structural necessity. In this framework, the 'strings' observed are not independent entities but are the physical manifestation of topological 'creases' formed where the local manifold and the shadow manifold intersect. The frame-dragging and geometric shear generated at the boundary of these nested manifolds naturally 'pinches' spacetime, creating localized lines of extreme tension. Consequently, the distinct 'butterfly' lensing profile observed in candidate CSc-1 serves as a direct, observational footprint of this manifold intersection.

C.2 Visualizations of Cosmic Anisotropy and Axis Orientations

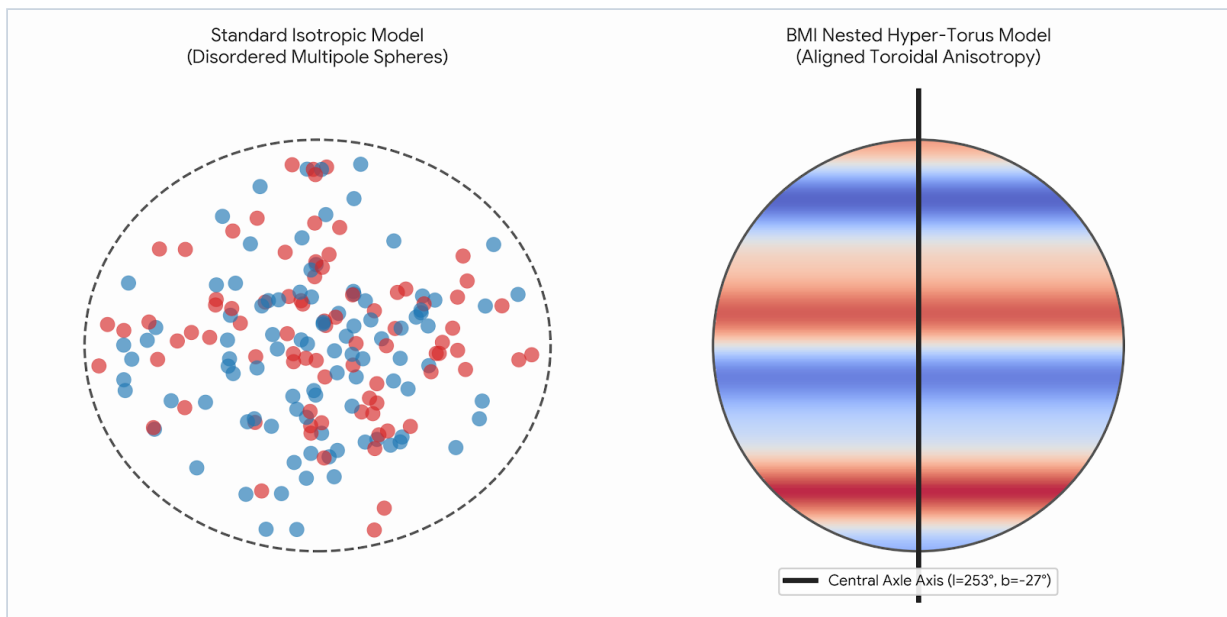


Figure C.1: Anisotropy Projections. Comparison between the Standard Isotropic Model featuring disordered multipole distributions and the BMI Aligned Toroidal Anisotropy Model organized along the Central Axle Axis ($l=253^\circ$, $b=-27^\circ$).

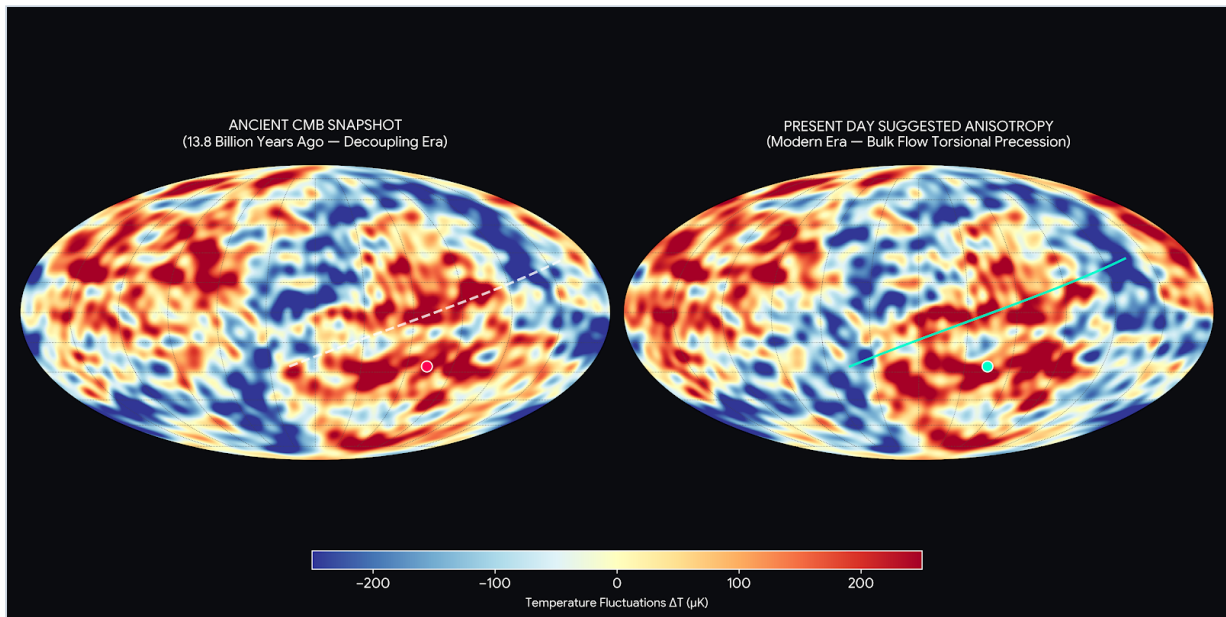


Figure C.2: Cosmic Microwave Background Evolution. Comparison of the Ancient CMB Snapshot at the Decoupling Era versus the Present Day Suggested Anisotropy map showing bulk flow torsional precession vectors.